Subject: JUQ manuscript #M173361
Date: Monday, 23 June 2025 at 4:32:07 PM New Zealand Standard Time
From: juq@siam.org
To: Oliver Maclaren
CC: o.maclaren@gmail.com

Dear Oliver Maclaren,

Your manuscript, "Invariant Image Reparameterisation: A Unified Approach to Structural and Practical Identifiability and Model Reduction," has completed this round of review.
Though I can't accept it for publication at this time, I encourage you to read the referee reports and revise the paper following the suggestions outlined.

Please carefully address every comment made by the reviewers and the Associate Editor, and respond to each accordingly. Your revision should also include a response to the referees with detailed information on the actions taken in view of the reports.

If you do not intend to make changes and send a revised version of the paper, please let me know by return email (with a cc: to juq@siam.org). Otherwise, I'm asking that you complete your revision within 2 months, by August 21, 2025. Be sure to let SIAM know (email both myself and juq@siam.org) if you require more time.

Note: A second email will follow, which will include a link that you, as the corresponding author, can use when you're ready to submit the revised version. Do not forward the follow-up email to anyone else as the link is for your use exclusively, and allows access to your account.

Best regards,

Bani Mallick, Sebastian Reich
Editors-in-Chief
SIAM/ASA Journal on Uncertainty Quantification

Associate Editor (Remarks to the Author):

Reading the referee reports I see that one of the major criticisms of the reviewers is the lack of a compelling computed example. Therefore, as the ideas in the paper are interesting,I would encourage to apply the methodology to a more ambitious problem.

Referee #1 (Remarks to the Author):

This paper presents a methodology called IIR, invariant image reparameterization. Its aim is to find identifiable parameter combinations and use them for model reduction, leading to reduced models with comparable predictive capabilities. It is a natural continuation of the authors' previous work on profile-wise analysis (PWA) and the use of unidentifiable models for making predictions. While the work is potentially interesting, it raises a number of concerns. The major ones are as follows:

1. In its current state, the description of the IIR methodology is somewhat ambiguous. It should be defined in an explicit and systematic way, and each of its steps should be clearly described. Maybe a diagram showing the method steps in an algorithmic fashion could be helpful.

2. The differences between IIR and another methodology presented in several papers by Stigter, Molenaar and others (see: https://doi.org/10.1016/j.mbs.2020.108328, https://doi.org/10.1007/s11071-021-07125-4, and

related works) are unclear. Both methods rely on a combination of symbolic and numerical computations, with a singular value decomposition of a sensitivity-related matrix. Given their similarities, it is perhaps surprising that these works are not cited in the submitted paper. The authors should discuss the similarities and the differences between both approaches, and compare them critically from theoretical and practical perspectives, justifying the advantages of the new method.

3. The fact that the authors have illustrated their methodology with three case studies of different types is a valuable aspect of the paper. However, all of them are very small, almost toy models. It is fine to use such models to explain the details of the method, but it would be desirable to show how the method performs with larger, more realistic problems.

4. It is difficult to grasp the advantages of using IIR for uncertainty quantification from the results shown in the manuscript, particularly the SM1 and SM2 figures of the Supplementary Material. One would imagine that this issue would only get worse in models with more parameters, which is one reason why it would be helpful to include such a case study.

Minor comments:

1. The expressions "Practically non-identified" and "structurally non-identified" are used already in the Abstract, and repeated much later in the paper, but they are not defined. These are not standard terms (unlike e.g. practically unidentifiable) and their meaning is not obvious, so they should be defined.

2. From the author contributions stated in the paper, it seems that only the first and last authors should be on the author list. Contributing to review and editing of the manuscript does not qualify for authorship.

Referee #2 (Remarks to the Author):

The authors consider the problem of non-identifiability in parameter estimation problems, introducing a novel approach that they call the Invariant Image Reparametrization (IIR). In this approach, the mapping from the model parameters to data distribution parameters is factored as a combination of a surjective mapping to identifiable parameters, and a one-to-one immersion to data distribution parameters. ("epi-mono-factorization"), on which the identifiability problem is analyzed.

My main criticism of this article is that the abstract theory in Chapter 2 is developed on a very general level that leaves the reader to only guess what kind of practical problems indeed can be solved in this framework. To be more specific, the starting point for the discussion is the mapping from "model parameters" to "data distribution parameters", but the difference between these two parameter types is not explained. The epi-mono-factorization is described through an "image space" in category-theoretical way, but it is not explained for what types of models such factorization is guaranteed. The models are then rewritten for logarithmic parameters, and it is stated that "models that can be written in terms of monomial parameter combinations take the form (2.10)". What does the concept "monomial parameter" refer to, and what types of models allow such parametrization is not clear. This seems to be a statement that limits somehow the formalism to particular types of models. For the subsequent development of the theory, however, the specific form (2.10) seems to be critical, so it seems quite crucial to specify the models. Also, it is not clear how and why the mapping A in (2.10) can be identified with the singular vector matrix in (2.13). The discussion after formula (2.14) gives an impression that the basic ideas are after all much simpler than what the general abstract theory suggests (it looks like the identifiable parameters correspond to non-zero singular values), so it left me wonder if all the elaborate category theoretic machinery is indeed necessary or useful. Finally, I was hoping to find an answer to my questions by following through the computed examples, but to my disappointment, the paper contains only very elementary low-dimensional examples that can be analyzed without much sophistication anyway, so I was left to wonder how useful all the machinery in the paper indeed is.

My criticism does not imply that the paper does not have its merits, and I believe that if the concepts had been elucidated by non-trivial examples in a less top down manner where all the theoretical work comes into display, the article could be of interest to the readership of this journal. However, in its present form, I cannot recommend its publication.